three hours in detecting an aerator failure can result in the loss of an entire pond. AquaExchange's IoT system continuously monitors aerator operations and sends an alert by phone if an aerator stops working for more than 10 minutes.

The company also aims to reduce electricity costs, which are the second-largest expense in shrimp farming. Its power management system monitors energy consumption and improves electrical efficiency by raising the power factor to around 0.98, helping farmers lower their electricity bills.

Another solution is an IoT-enabled robotic feeder, a solar-powered device that automates shrimp feeding. "We have developed an AI-based feeding system that uses sensitive underwater microphones to capture the sounds shrimp make while eating. The AI analyses these acoustic signals to estimate the shrimp's hunger levels and adjusts feeding accordingly. This helps improve feed efficiency by around 10% while reducing feed wastage," Pavan says.

The company has also developed AI-powered health monitoring tools that analyse microscopic images of shrimp organs, including the hepatopancreas and gut, as well as water samples. Using custom AI models, the system can detect stress and predict disease risks three to four days before visible symptoms appear. According to the startup, the system helps reduce mortality losses by 40-50 per cent.



Pavan Kosaraju, Founder and CEO, AquaExchange

Speaking about the design challenges, Pavan adds, "Our products operate in nearly 100% humidity, temperatures ranging from 6-7°C to over 48°C, with constant water splashes, feed dust, and continuous vibrations. We also have to account for rough handling in the field, unlike controlled industrial environments. To withstand these conditions, we designed our devices with IP65-rated protection. Even then, vibration remained one of our biggest engineering challenges, leading us to carry out detailed acoustic and vibration analysis before finalising every product design."

On the company's financial performance, Pavan Kosaraju says, "We closed the financial year ended March 2026 with revenue of around ₹2.75 billion. We have also reached operational break-even, so our core business is now financially sustainable."

Speaking about the company's current challenges, Pavan says, "Our biggest challenge today is reliability. As we scale, we have to ensure not just that our technology works consistently, but also that our manufacturing processes, supply chains, and operations remain reliable.

Looking ahead, the company has three main growth priorities. In the shrimp sector, it aims to increase its market share by around 10% each year while increasing adoption of its technology among existing customers. The second is international expansion, where the company says it is already among the top two players. The third is expanding beyond shrimp by applying its technology to other aquaculture species.

—Nidhi Agarwal

3 COMPCARTA REDUCES THE COST AND COMPLEXITY OF END-TO-END CHIP DESIGN

Building a chip should not require assembling an entire design ecosystem. A NOIDA-based startup is developing a browser-based EDA platform that aims to reduce dependence on expensive software and high-performance computing, making semiconductor design more accessible to startups, researchers, and emerging chip companies.

Compcarta, a Greater NOIDA-based semiconductor startup, has built an end-to-end EDA (electronic design automation) platform aimed at simplifying chip design by addressing high software costs, complex toolchains, demanding computing requirements, and limited access to professional design tools. Designed for enterprises, startups, early-stage design companies, and academic institutions, the platform combines cloud infrastructure or on-premises deployment with a mix of open source and proprietary EDA technologies to reduce the cost and complexity of deploying end-to-end chip design flows.

The company's EDAcloud platform is also hosted on the National EDA Tool Grid under the ChipIN initiative, making its tools accessible to academic institutions across India. Since its deployment, the platform has attracted nearly 700 academic users and is being explored by 10 startups,